

Supplementary materials

Table of contents

Supplementary Table 1. Descriptive statistics of PM _{2.5} exposures, hospitalization costs, and length of stay stratified by regions in Brazil, 2000–2019.	2
Supplementary Table 2. Characteristics of wildfire-specific PM _{2.5} and meteorological factors in the included locations, by Brazilian regions.	3
Supplementary Table 3. Attributable losses and fractions of hospitalization costs and length of stay from all-cause, respiratory, and cardiovascular diseases to wildfire-specific PM _{2.5} from 2000 to 2019, by sex and age.	4
Supplementary Table 4. Attributable losses and fractions of hospitalization costs and length of stay from cause-specific respiratory diseases to wildfire-specific PM _{2.5} from 2000 to 2019, by sex and age.	5
Supplementary Table 5. Attributable losses and fractions of hospitalization costs and length of stay from cause-specific cardiovascular diseases to wildfire-specific PM _{2.5} from 2000 to 2019, by sex and age.	6
Supplementary Figure 1. First-stage and reduced-form diagnostic binscatter plots. . .	7
Supplementary Figure 2. Spatiotemporal patterns of wildfire-specific PM _{2.5} , non-wildfire PM _{2.5} , and wind speed in Brazil, 2000–2019.	8
Supplementary Figure 3. Distribution of relative underestimation by traditional analysis.	9
Supplementary Figure 4. Comparison of estimated percentage changes associated with wildfire-specific and non-wildfire PM _{2.5}	10
Supplementary Figure 5. Sensitivity analyses of the association between wildfire-specific PM _{2.5} and hospitalization outcomes.	11

Supplementary Table 1. Descriptive statistics of PM_{2.5} exposures, hospitalization costs, and length of stay stratified by regions in Brazil, 2000–2019.

Variables	Nationwide	North	Northeast	Southeast	South	Central-west
No. of locations	449	18	148	137	95	51
Wildfire-specific PM _{2.5} (Median [IQR], µg/m ³)	1.7 (3.1)	1.6 (4.8)	1.0 (2.1)	2.3 (2.8)	3.8 (7.3)	1.6 (2.9)
Other-sourced PM _{2.5} (Median [IQR], µg/m ³)	11.7 (4.4)	12.0 (5.3)	12.1 (4.2)	12.2 (4.5)	10.8 (3.9)	10.4 (5.3)
Days with zero admissions (%)	17843 (0.5)	1157 (0.9)	5351 (0.5)	1263 (0.1)	6840 (1.0)	3232 (0.9)
Hospitalization costs per admission (Median [IQR], \$)						
All-cause diseases	295.7 (267.0)	205.0 (160.7)	251.7 (201.6)	346.9 (310.6)	357.8 (327.6)	258.0 (223.1)
Respiratory diseases	238.3 (210.4)	215.6 (195.0)	212.6 (191.6)	262.2 (236.0)	256.6 (213.8)	217.7 (197.5)
Cardiovascular diseases	360.4 (614.3)	261.7 (346.9)	277.1 (355.9)	439.7 (704.7)	477.0 (792.1)	308.5 (521.7)
Length of stay per admission (Median [IQR], days)						
All-cause diseases	4.3 (2.5)	3.7 (2.3)	3.8 (2.3)	4.7 (2.5)	4.7 (2.6)	3.8 (2.3)
Respiratory diseases	4.0 (3.0)	4.0 (3.6)	3.8 (2.4)	4.6 (3.2)	4.3 (2.8)	3.6 (2.1)
Cardiovascular diseases	4.5 (3.7)	4.8 (5.0)	4.2 (3.7)	4.8 (3.5)	4.5 (3.5)	4.0 (3.1)

Note: Data represents statistics for the 449 immediate regions included in the analysis over the study period.

Abbreviations: IQR, interquartile range. Days with zero admissions refers to the number (percentage) of region-days with zero hospitalization events for all-cause diseases. Hospitalization costs and length of stay are presented as median (IQR) of the daily average values per admission in all regions.

Supplementary Table 2. Characteristics of wildfire-specific PM_{2.5} and meteorological factors in the included locations, by Brazilian regions.

Variables	Mean (SD)	Minimum	Median	Maximum	Interquartile Range
Wildfire-specific PM _{2.5} (µg/m ³)					
Overall	3.3 (4.5)	0	1.7	63.3	3.1
North	4.9 (7.8)	0	1.6	63.3	4.8
Northeast	2.1 (3.0)	0	1	37.1	2.1
Central-West	2.9 (3.7)	0	1.6	39.7	2.9
Southeast	3.3 (3.3)	0	2.3	45.9	2.8
South	6.8 (7.5)	0	3.8	63.2	7.3
Wind speed (m/s)					
Overall	2.3 (1.3)	0	2.1	11.6	1.8
North	1.4 (0.8)	0	1.2	5.2	1.1
Northeast	3.0 (1.3)	0	3	10.1	1.9
Central-West	1.8 (1.0)	0	1.7	9.2	1.3
Southeast	2.2 (1.2)	0	2.1	11.6	1.7
South	1.7 (0.9)	0	1.6	7.2	1.2
Temperature (°C)					
Overall	23.4 (4.1)	-0.6	26.3	35.4	5.1
North	26.1 (1.6)	12.6	26.1	32.2	1.9
Northeast	26.1 (2.2)	12	26.2	34	2.9
Central-West	22.3 (3.2)	6.3	22.5	35.4	4.4
Southeast	19.5 (4.7)	-0.6	20.1	33.3	6.4
South	24.7 (2.7)	4	25	34.5	3
Relative humidity (%)					
Overall	72.1 (13.2)	16.4	74.3	187.5	18.1
North	82.3 (9.8)	30.6	85	187.5	10.1
Northeast	70.2 (13.1)	21.5	72.5	172.9	19.4
Central-West	71.0 (12.7)	19	72.6	169.9	17
Southeast	77.0 (10.5)	24.3	78	175.2	14.6
South	68.2 (15.6)	16.4	71	171.1	24.4

Abbreviations: SD: standard deviation.

Supplementary Table 3. Attributable losses and fractions of hospitalization costs and length of stay from all-cause, respiratory, and cardiovascular diseases to wildfire-specific PM_{2.5} from 2000 to 2019, by sex and age.

Characteristics	All-cause diseases		All-cause respiratory diseases		All-cause cardiovascular diseases	
	Attributable loss	Attributable fraction	Attributable loss	Attributable fraction	Attributable loss	Attributable fraction
Hospitalization costs (million US\$, 95% CI)						
Total	755.60 (468.72, 1039.56)	0.96 (0.59, 1.32)	331.79 (275.58, 386.90)	4.20 (3.49, 4.90)	109.95 (8.38, 209.80)	0.69 (0.05, 1.32)
Sex						
Female	383.27 (237.75, 527.31)	0.95 (0.59, 1.31)	151.63 (125.94, 176.82)	4.19 (3.48, 4.89)	48.06 (3.66, 91.71)	0.69 (0.05, 1.31)
Male	372.33 (230.97, 512.25)	0.97 (0.60, 1.33)	180.16 (149.65, 210.08)	4.21 (3.50, 4.91)	61.89 (4.72, 118.09)	0.69 (0.05, 1.33)
Age group, years						
0-19	158.29 (98.20, 217.75)	0.95 (0.59, 1.31)	126.94 (105.42, 148.05)	4.00 (3.32, 4.67)	2.18 (0.17, 4.15)	0.68 (0.05, 1.30)
20-59	364.98 (226.40, 502.14)	0.95 (0.59, 1.30)	85.65 (71.14, 99.87)	4.25 (3.53, 4.96)	44.16 (3.36, 84.27)	0.69 (0.05, 1.31)
≥60	232.34 (144.12, 319.67)	0.98 (0.61, 1.35)	119.20 (99.02, 138.97)	4.40 (3.65, 5.13)	63.61 (4.85, 121.38)	0.69 (0.05, 1.32)
Length of stay (million days, 95% CI)						
Total	30.75 (23.72, 37.71)	0.97 (0.74, 1.18)	5.39 (4.73, 6.05)	4.30 (3.77, 4.83)	2.91 (2.16, 3.65)	1.25 (0.93, 1.57)
Sex						
Female	13.73 (10.59, 16.84)	1.00 (0.77, 1.23)	2.47 (2.17, 2.78)	4.35 (3.82, 4.88)	1.35 (1.00, 1.69)	1.26 (0.93, 1.58)
Male	17.02 (13.13, 20.88)	0.94 (0.72, 1.15)	2.92 (2.56, 3.27)	4.26 (3.73, 4.78)	1.56 (1.16, 1.96)	1.24 (0.92, 1.56)
Age group, years						
0-19	4.22 (3.26, 5.18)	1.42 (1.10, 1.74)	2.06 (1.81, 2.32)	4.21 (3.69, 4.72)	0.05 (0.04, 0.07)	1.66 (1.24, 2.08)
20-59	18.63 (14.37, 22.86)	0.89 (0.69, 1.09)	1.34 (1.18, 1.50)	4.23 (3.70, 4.74)	1.04 (0.77, 1.30)	1.28 (0.95, 1.60)
≥60	7.89 (6.09, 9.68)	1.00 (0.77, 1.22)	1.99 (1.74, 2.23)	4.46 (3.91, 5.00)	1.82 (1.35, 2.29)	1.23 (0.91, 1.54)

Note: All the economic costs of different year have been adjusted to 2019 U.S. dollars levels using the consumer price index.

Supplementary Table 4. Attributable losses and fractions of hospitalization costs and length of stay from cause-specific respiratory diseases to wildfire-specific PM_{2.5} from 2000 to 2019, by sex and age.

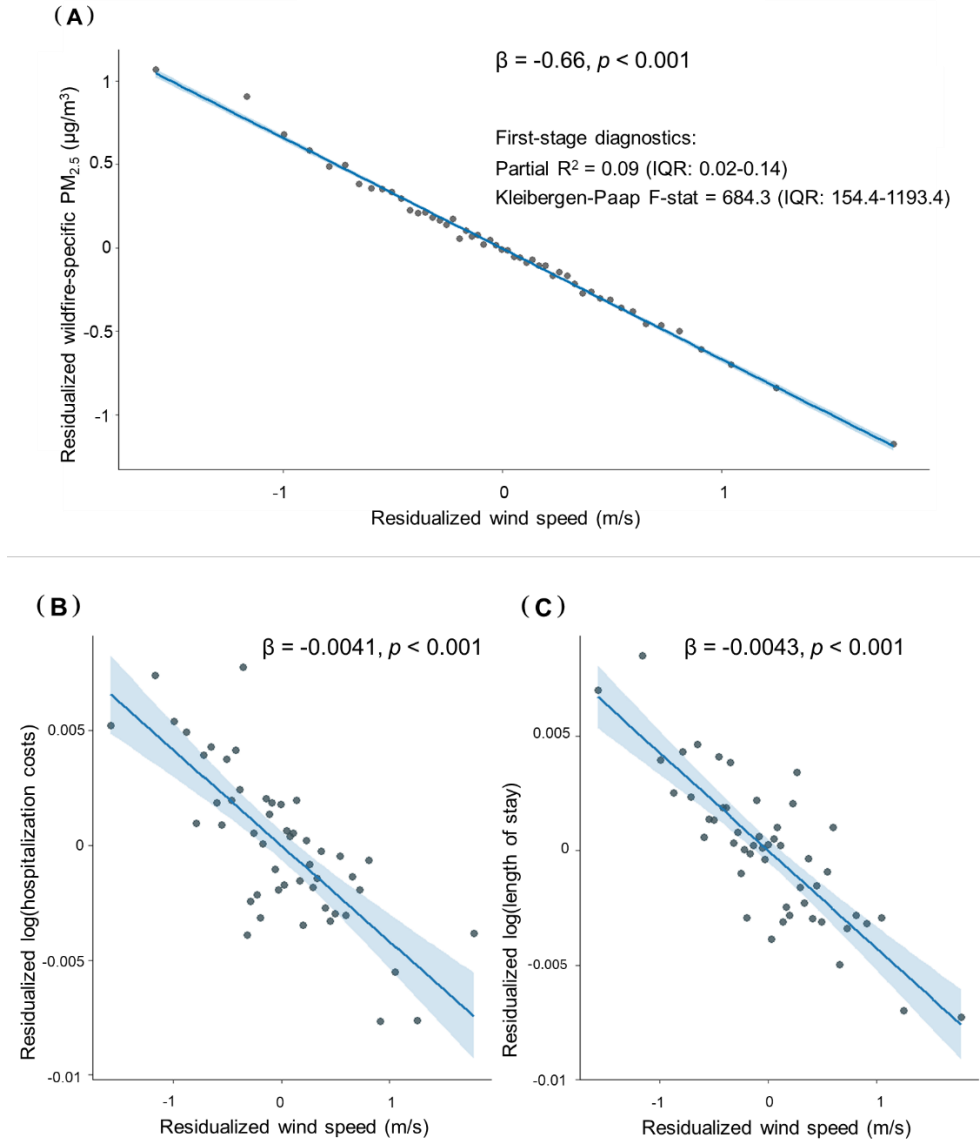
Characteristics	Pneumonia		COPD		Asthma	
	Attributable loss	Attributable fraction	Attributable loss	Attributable fraction	Attributable loss	Attributable fraction
Hospitalization costs (million US\$, 95% CI)						
Total	177.41 (145.79, 208.31)	4.63 (3.80, 5.43)	34.92 (26.74, 42.87)	4.70 (3.60, 5.77)	39.20 (33.31, 44.93)	6.02 (5.12, 6.90)
Sex						
Female	82.26 (67.60, 96.59)	4.62 (3.80, 5.42)	16.29 (12.47, 20.00)	4.73 (3.62, 5.81)	19.99 (16.99, 22.91)	6.09 (5.18, 6.98)
Male	95.15 (78.19, 111.72)	4.64 (3.81, 5.44)	18.63 (14.27, 22.87)	4.67 (3.58, 5.74)	19.21 (16.32, 22.02)	5.95 (5.06, 6.82)
Age group, years						
0-19	77.73 (63.85, 91.29)	4.35 (3.57, 5.11)	2.09 (1.60, 2.57)	4.19 (3.21, 5.15)	24.16 (20.51, 27.71)	5.69 (4.84, 6.53)
20-59	38.77 (31.87, 45.50)	4.83 (3.97, 5.66)	7.66 (5.87, 9.41)	4.42 (3.38, 5.43)	9.17 (7.80, 10.50)	6.63 (5.64, 7.59)
≥60	60.92 (50.07, 71.51)	4.90 (4.03, 5.76)	25.17 (19.28, 30.88)	4.84 (3.71, 5.94)	5.87 (5.00, 6.72)	6.65 (5.66, 7.61)
Length of stay (million days, 95% CI)						
Total	3.05 (2.63, 3.46)	4.69 (4.04, 5.33)	0.75 (0.59, 0.91)	3.95 (3.11, 4.77)	0.56 (0.47, 0.65)	5.73 (4.79, 6.65)
Sex						
Female	1.41 (1.21, 1.60)	4.68 (4.04, 5.32)	0.34 (0.27, 0.41)	4.31 (3.39, 5.20)	0.29 (0.24, 0.34)	5.80 (4.84, 6.72)
Male	1.64 (1.41, 1.86)	4.70 (4.05, 5.33)	0.41 (0.32, 0.50)	3.69 (2.90, 4.46)	0.27 (0.22, 0.31)	5.66 (4.73, 6.57)
Age group, years						
0-19	1.33 (1.15, 1.51)	4.45 (3.84, 5.06)	0.03 (0.02, 0.04)	4.08 (3.21, 4.93)	0.33 (0.28, 0.38)	5.43 (4.54, 6.30)
20-59	0.67 (0.58, 0.76)	4.80 (4.14, 5.45)	0.18 (0.14, 0.22)	3.33 (2.62, 4.03)	0.14 (0.11, 0.16)	6.23 (5.21, 7.21)
≥60	1.04 (0.90, 1.19)	4.96 (4.27, 5.63)	0.54 (0.42, 0.65)	4.21 (3.31, 5.08)	0.09 (0.08, 0.11)	6.24 (5.22, 7.23)

Note: All the economic costs of different year have been adjusted to 2019 U.S. dollars levels using the consumer price index.

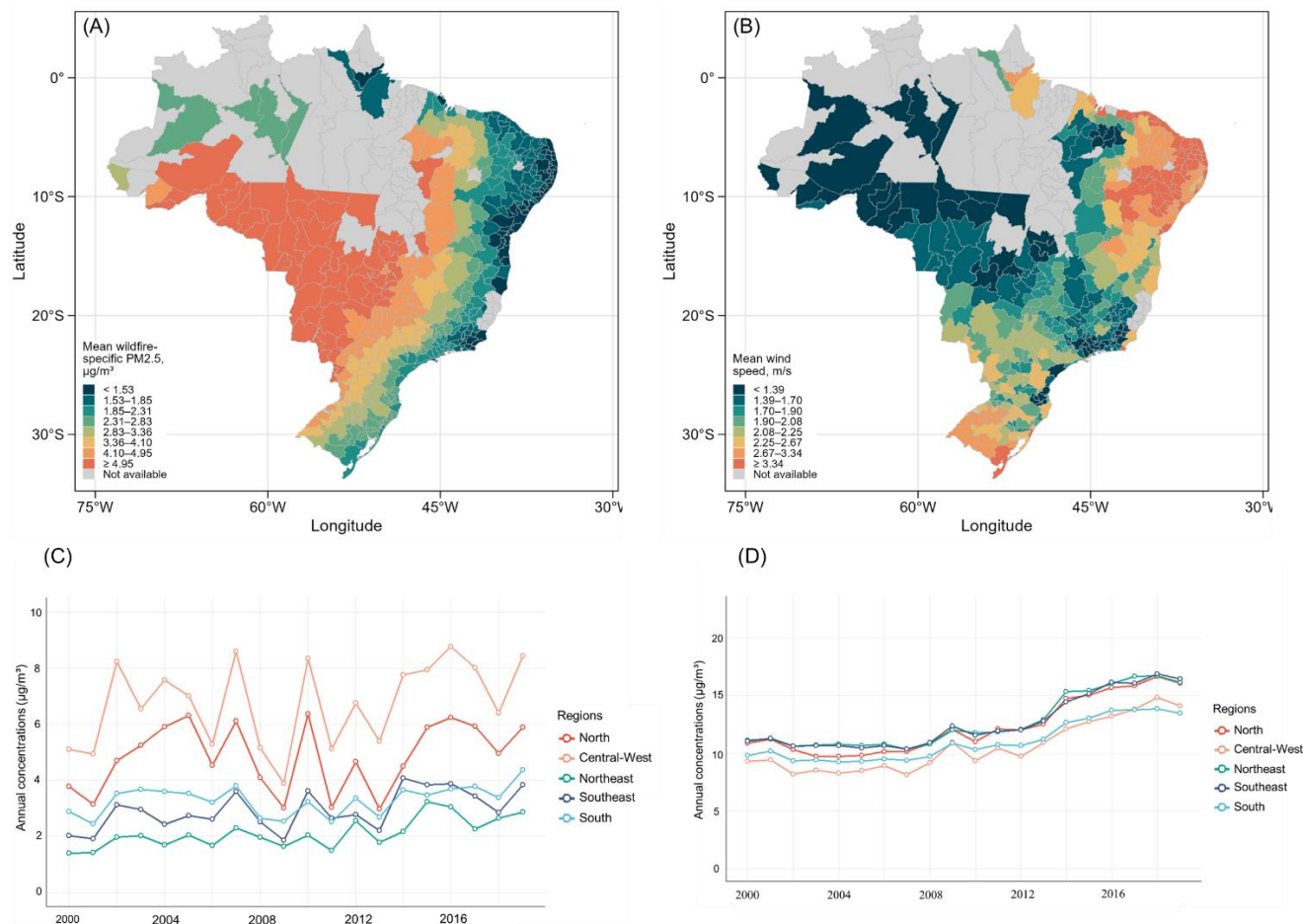
Supplementary Table 5. Attributable losses and fractions of hospitalization costs and length of stay from cause-specific cardiovascular diseases to wildfire-specific PM_{2.5} from 2000 to 2019, by sex and age.

Characteristics	Ischemic heart disease		Heart failure		Stroke	
	Attributable loss	Attributable fraction	Attributable loss	Attributable fraction	Attributable loss	Attributable fraction
Hospitalization costs (million US\$, 95% CI)						
Total	19.18 (-40.75, 77.57)	0.32 (-0.69, 1.31)	103.49 (82.57, 123.87)	4.76 (3.80, 5.70)	-23.43 (-53.42, 5.32)	-1.24 (-2.83, 0.28)
Sex						
Female	6.87 (-14.60, 27.79)	0.32 (-0.68, 1.30)	49.62 (39.58, 59.41)	4.75 (3.79, 5.69)	-11.89 (-27.10, 2.70)	-1.22 (-2.79, 0.28)
Male	12.31 (-26.15, 49.78)	0.32 (-0.69, 1.31)	53.87 (42.99, 64.47)	4.77 (3.81, 5.71)	-11.54 (-26.32, 2.62)	-1.26 (-2.87, 0.29)
Age group, years						
0-19	0.03 (-0.06, 0.11)	0.33 (-0.70, 1.33)	3.00 (2.40, 3.59)	4.68 (3.73, 5.60)	-0.36 (-0.81, 0.08)	-1.24 (-2.81, 0.28)
20-59	7.68 (-16.33, 31.08)	0.32 (-0.68, 1.30)	30.59 (24.40, 36.61)	4.64 (3.70, 5.55)	-9.39 (-21.40, 2.13)	-1.23 (-2.80, 0.28)
≥60	11.47 (-24.36, 46.38)	0.32 (-0.69, 1.31)	69.90 (55.77, 83.67)	4.82 (3.85, 5.77)	-13.69 (-31.21, 3.11)	-1.25 (-2.85, 0.28)
Length of stay (million days, 95% CI)						
Total	0.34 (0.12, 0.55)	1.26 (0.44, 2.06)	1.30 (1.07, 1.54)	4.47 (3.66, 5.26)	-0.15 (-0.43, 0.12)	-0.59 (-1.69, 0.47)
Sex						
Female	0.13 (0.05, 0.22)	1.24 (0.43, 2.03)	0.63 (0.52, 0.75)	4.47 (3.66, 5.26)	-0.07 (-0.21, 0.06)	-0.59 (-1.67, 0.47)
Male	0.20 (0.07, 0.33)	1.27 (0.44, 2.08)	0.67 (0.55, 0.79)	4.47 (3.66, 5.25)	-0.08 (-0.22, 0.06)	-0.60 (-1.71, 0.48)
Age group, years						
0-19	0.00 (0.00, 0.00)	1.33 (0.46, 2.18)	0.03 (0.02, 0.03)	4.34 (3.55, 5.10)	-0.00 (-0.00, 0.00)	-0.62 (-1.77, 0.49)
20-59	0.13 (0.05, 0.22)	1.27 (0.44, 2.07)	0.38 (0.31, 0.45)	4.30 (3.52, 5.06)	-0.05 (-0.14, 0.04)	-0.59 (-1.68, 0.47)
≥60	0.20 (0.07, 0.33)	1.26 (0.44, 2.06)	0.90 (0.73, 1.05)	4.54 (3.72, 5.35)	-0.10 (-0.29, 0.08)	-0.59 (-1.69, 0.47)

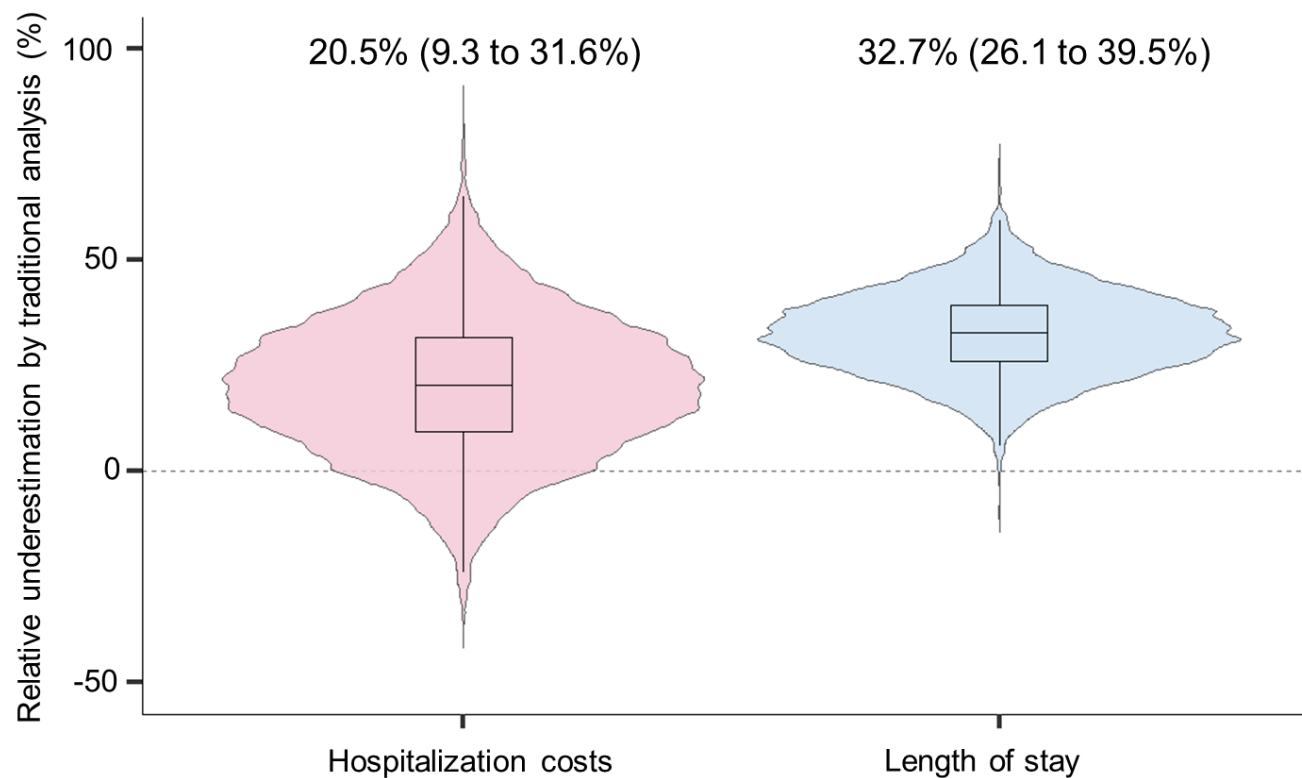
Note: All the economic costs of different year have been adjusted to 2019 U.S. dollars levels using the consumer price index.



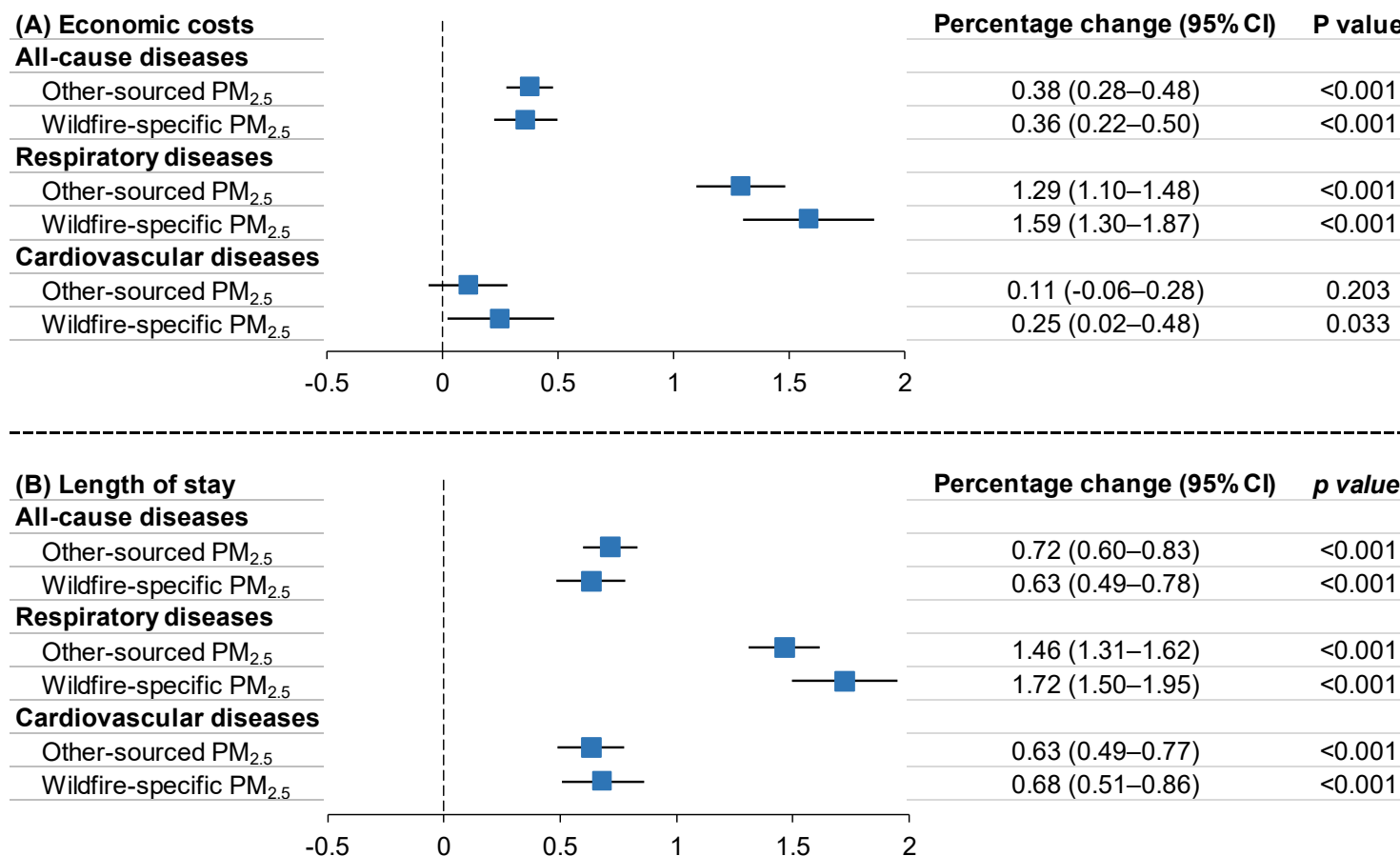
Supplementary Figure 1. First-stage and reduced-form diagnostic binscatter plots. Panel (A) displays a binned scatterplot of the residuals of daily wildfire-specific $PM_{2.5}$ concentrations against the residuals of daily wind speed. The annotations indicate the median and interquartile range (IQR) values of the Kleibergen-Paap F-statistic and Partial R^2 across all immediate regions in the first stage. Panels (B-C) show reduced-form relationships of the residuals of log-transformed hospitalization costs (B) and length of stay (C) against the residuals of daily wind speed. In all panels, the x-axis was divided into 50 equal-sized bins based on the distribution of residualized wind speed. The gray dots represent the mean values of the residuals within each bin, the blue lines represent the linear regression fits, and the shaded areas indicate 95% confidence intervals. The term residualized or residuals indicates that all plotted variables are the exact residuals obtained after partialling out high-dimensional space-time fixed effects and covariates (temperature, relative humidity, holiday, and non-wildfire $PM_{2.5}$).



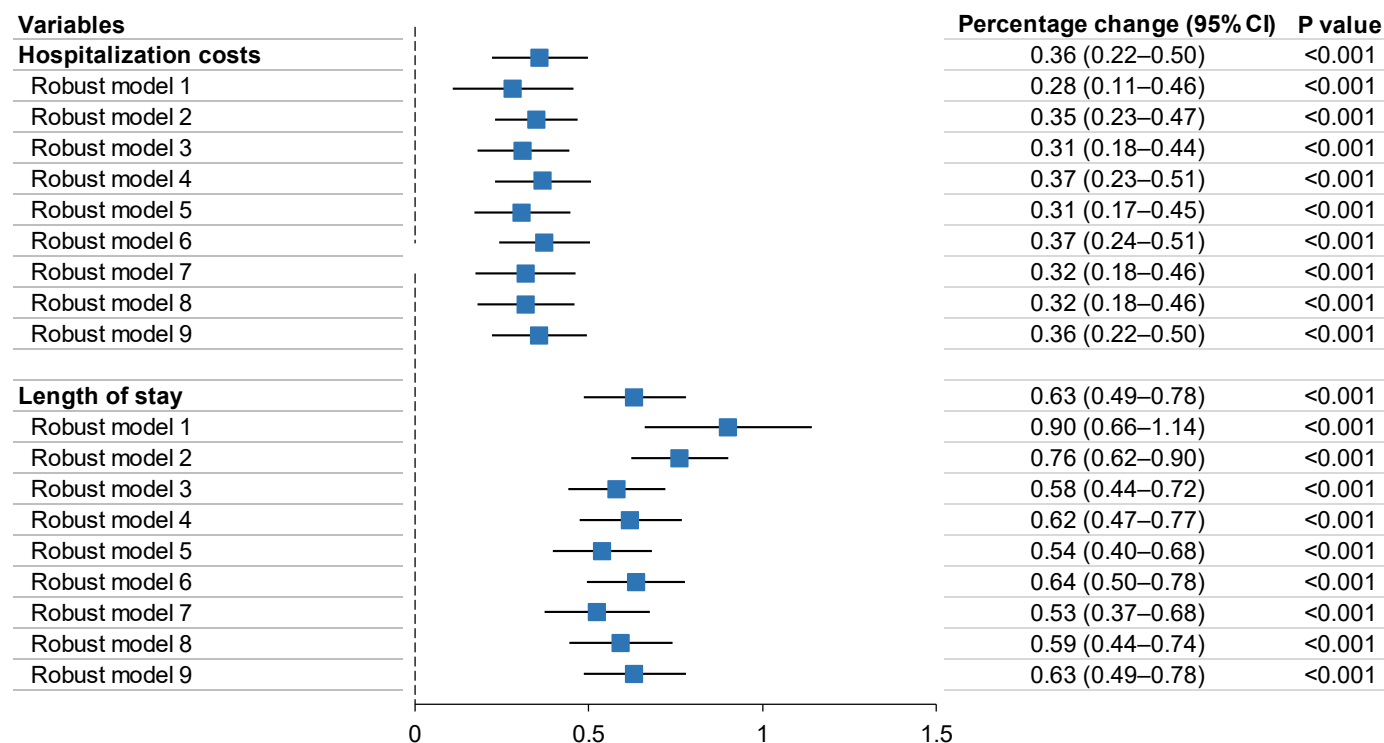
Supplementary Figure 2. Spatiotemporal patterns of wildfire-specific PM_{2.5}, non-wildfire PM_{2.5}, and wind speed in Brazil, 2000–2019. (A) Spatial distribution of the 20-year average wildfire-specific PM_{2.5} concentrations across 449 immediate regions. (B) Spatial distribution of the 20-year average wind speed (m/s). (C) Annual trends in wildfire-specific PM_{2.5} concentrations stratified by five macro-regions (North, Northeast, Central-West, Southeast, and South). (D) Annual trends in non-wildfire PM_{2.5} concentrations by region.



Supplementary Figure 3. Distribution of relative underestimation by traditional analysis. The figure displays the distribution of relative underestimation of wildfire-specific PM_{2.5} effects on hospitalization costs (left) and length of stay (right). The relative underestimation was calculated as the percentage difference between the causal and traditional estimates based on Monte Carlo simulations (n=10,000). The annotated values denote the median percentage underestimation with the interquartile range (IQR) values.



Supplementary Figure 4. Comparison of estimated percentage changes associated with wildfire-specific and non-wildfire PM_{2.5}. Estimated percentage changes in (A) hospitalization costs and (B) length of stay per 1 µg/m³ increase in wildfire-specific or non-wildfire PM_{2.5} for all-cause, respiratory and cardiovascular diseases. Squares indicate point estimates and horizontal lines indicate 95% confidence intervals. The vertical dashed line indicates the null value. P values were obtained from the corresponding meta-analysis. n = 3,279,945 immediate-region days.



Supplementary Figure 5. Sensitivity analyses of the association between wildfire-specific PM_{2.5} and hospitalization outcomes.

The figure displays the percentage change and 95% confidence intervals in hospitalization costs and length of stay associated with a 1 µg/m³ increase in wildfire-specific PM_{2.5} across different model specifications. Robust model 1: using 16-sector wind direction as the instrumental variable (IV) instead of wind speed; Robust model 2: using both wind speed and wind direction as IVs; Robust model 3: excluding temperature and relative humidity from the covariates; Robust model 4: adjusting for temperature and relative humidity using a 5-day moving average window; Robust model 5: adjusting for temperature and relative humidity using a 3-day moving average window; Robust model 6: using a current-day (1-day) moving window for wildfire-specific PM_{2.5} and wind speed; Robust model 7: using a 3-day moving average window for wildfire-specific PM_{2.5} and wind speed; Robust model 8: excluding observations with wind speeds exceeding 5.4 m/s (optimal walking conditions); Robust model 9: excluding observations with wind speeds exceeding 8.7 m/s (Beaufort scale level 7).